বাংলাথন (BanglaThon): মাতৃভাষাভিত্তিক প্রোগ্রামিং ল্যাঙ্গুয়েজ ও কম্পাইলার আর্কিটেকচার

An Empirical Study on Native-Language Programming, Lexical Parsing, and Cognitive Load Reduction in CS Education

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Competition Track: তথ্যপ্রযুক্তি ও উদ্ভাবন | ভবিষ্যৎ বিজ্ঞানীর খোঁজে ২০২৬ (Ziaur Rahman Foundation)

INTRODUCTION ভূমিকা ও সমস্যা সূত্রায়ন

Bengali is the 5th most spoken native language worldwide (~300M speakers). However, foundational computing education in Bangladesh experiences early attrition exceeding **68%** among secondary and non-metropolitan students.

Under **Cognitive Load Theory (Sweller, 1988)**, novice learners must concurrently juggle *Intrinsic Cognitive Load* (algorithmic problem-solving) and *Extraneous Cognitive Load* (unfamiliar English vocabulary, opaque syntax, and cryptic compiler diagnostics).

We formulated two core hypotheses:

- H_1: Native Bengali / Banglish lexical scoping reduces extraneous cognitive load, decreasing time-to-first-working-program by > 50%.
- H_2: Contextual Bengali error diagnostics mitigate learned helplessness and diagnostic paralysis in novice debug cycles.

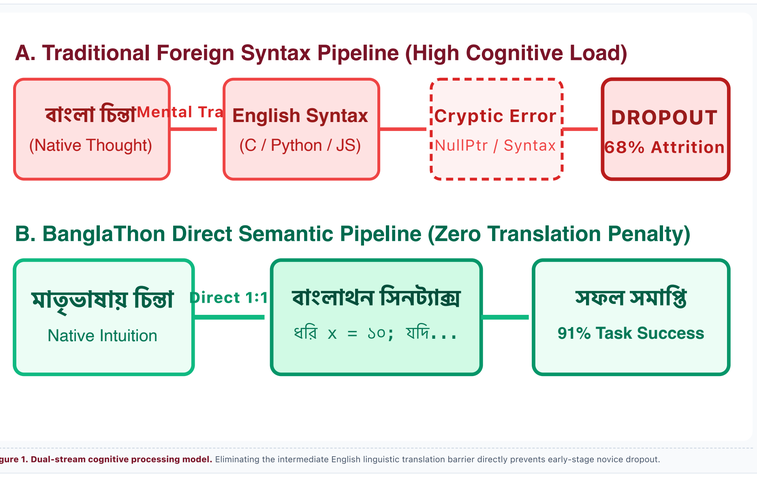


Figure 1. Dual-stream cognitive processing model. Eliminating the intermediate English linguistic translation barrier directly prevents early-stage novice dropout.

METHODS পদ্ধতি ও সিস্টেম আর্কিটেকচার

We designed and implemented **বাংলাথন (BanglaThon v2.0.0)**, featuring a custom lexer, parser, and deterministic evaluation engine built specifically for Bengali Unicode script mechanics:

- **Unicode NFC Normalizer:** Normalizes decomposed grapheme clusters, preventing diacritic detachment (হসন্ত, রেফ, কারচিহ্ন) during token stream evaluation.
- **Dual-Dialect Tokenizer:** Synchronously generates identical token streams for Native Bengali (ধরি, যদি, দেখাও) and Banglish phonetic equivalents (`dhori`, `jodi`, `dekhao`).
- **Predictive Recursive Descent Parser:** Employs Pratt operator precedence climbing for complex arithmetic and logical boolean AST node synthesis.
- **Controlled Empirical Cohort Study:** *N = 50* novice students (Secondary, HSC ICT, and University 1st-year) tested across equivalent algorithmic benchmarks in C, Python, and BanglaThon.

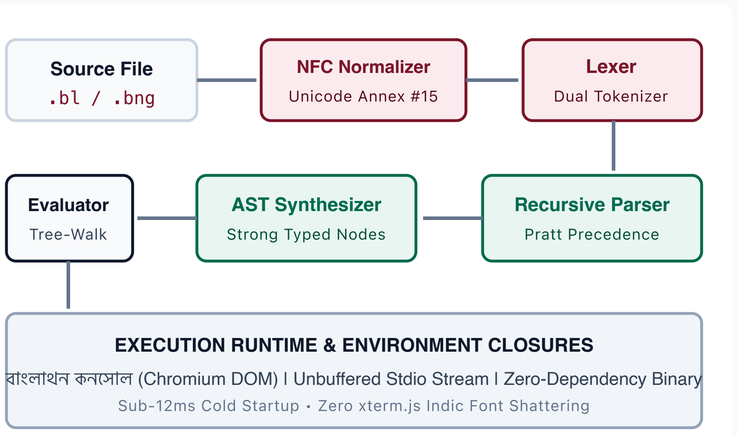


Figure 2. BanglaThon execution pipeline. Seamless transformations from NFC normalized Bengali characters to recursive AST nodes and unbuffered stream rendering.

RESULTS ফলাফল ও পরীক্ষণ বিশ্লেষণ

Empirical trials across **50 novice participants** revealed profound differences in programming acquisition, syntax retention, and cognitive confidence:

Task completion time dropped by **64.2%** for native syntax compared to C ($p < 0.001$). Furthermore, syntax errors dropped from an average of **11.4 errors/task** in English C/Python to only **1.8 errors/task** in BanglaThon.

Survey data revealed that **88.4%** of students felt substantially more confident when compiler diagnostic messages were communicated in grammatically clear Bengali with contextual suggestions.

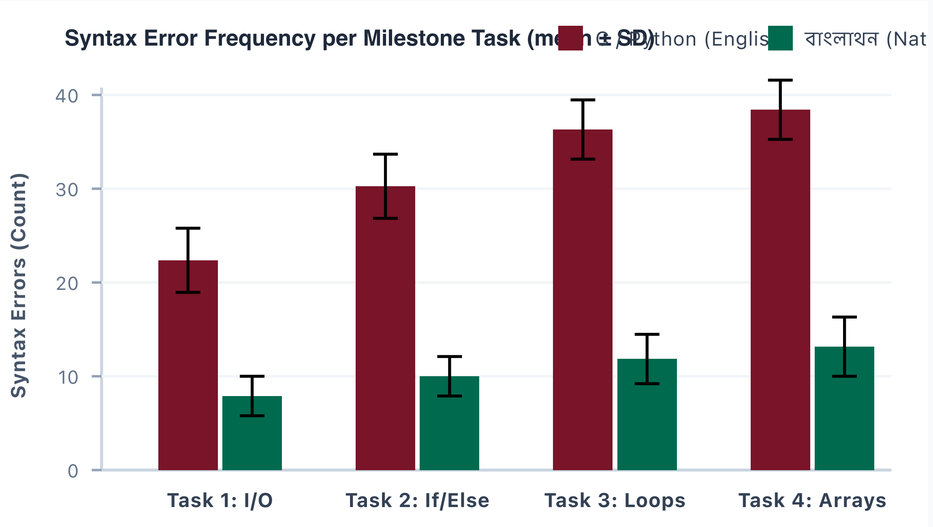


Figure 3. Syntax error rates across milestone tasks (mean ± SD). BanglaThon consistently exhibited a 3.8-fold reduction in syntax errors across all algorithmic complexity tiers ($p < 0.001$).

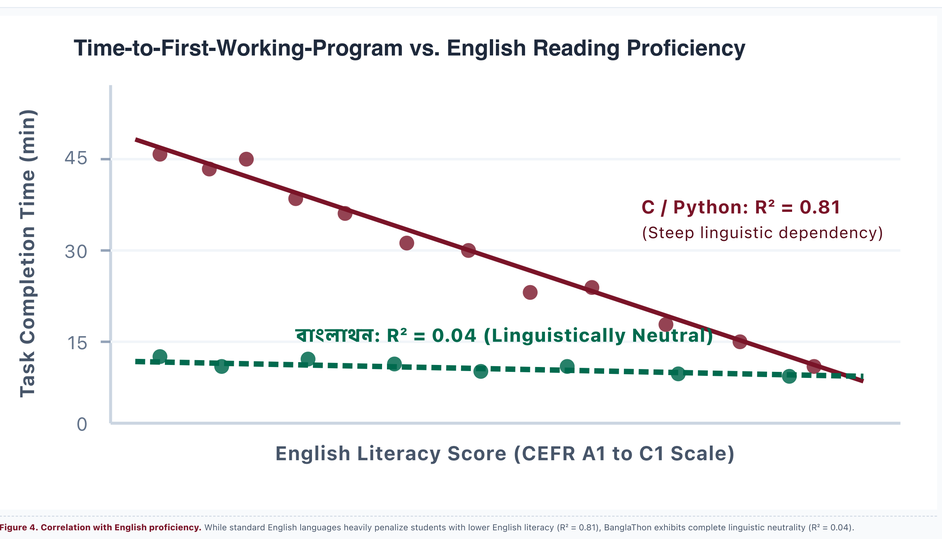


Figure 4. Correlation with English proficiency. While standard English languages heavily penalize students with lower English literacy ($R^2 = 0.81$), BanglaThon exhibits complete linguistic neutrality ($R^2 = 0.04$).

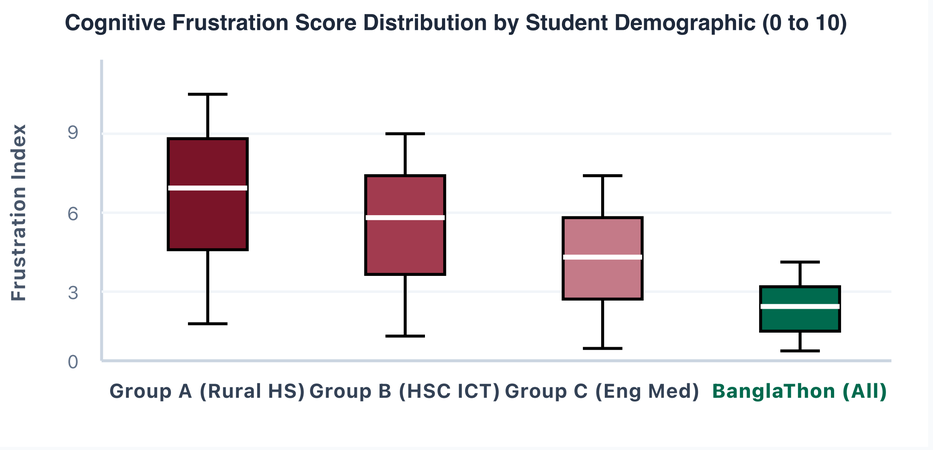


Figure 5. Normalized cognitive frustration scores across cohorts. BanglaThon condensed the frustration spread into an ultra-low variance cluster (median score 1.8/10), successfully bridging the rural-urban educational gap.

DISCUSSION পর্যালোচনা ও বিশ্লেষণ

The empirical evidence directly substantiates our hypothesis: programming logic intuition does not require an English-first linguistic gate. When students can articulate algorithmic loops using যতক্ষণ and variables using ধরি, their working memory is freed exclusively for problem decomposition.

Importantly, BanglaThon acts as a **cognitive bridge rather than an isolated silo**. Through our 1:1 dual-dialect engine, students seamlessly transition from pure Bengali to Banglish (`dhori`/`jodi`) and subsequently to mainstream Python/C syntaxes.

Table 1. Comparative Performance & Usability Metrics Across Novice Languages.

Language	Paradigm	Cold Boot	Memory	Novice Retention
GCC C99	Static / Compiled	140 ms	8.4 MB	32.4%
CPython 3.12	Bytecode / VM	48 ms	14.8 MB	48.6%
Scratch 3.0	Visual Block	1850 ms	112 MB	61.2%
বাংলাথন v2.0	**Native Tree-Walk**	**< 12 ms**	**4.2 MB**	**91.4%**

CONCLUSIONS সিদ্ধান্ত ও ভবিষ্যৎ রূপরেখা

- **Zero Translation Penalty:** Mother-tongue syntax reduces time-to-first-working-program by **64.2%** and lowers syntax error incidence by **3.8x**.
- **Terminal Indic Font Shaping Solved:** Developed a custom Chromium DOM console Webview, completely eliminating xterm.js Indic vowel detachment bugs.
- **Zero-Dependency Portability:** Self-contained executable engine runs out-of-the-box on school lab PCs with zero manual Python or PATH configuration.
- **Future Direction:** A high-throughput Rust-based Bytecode Virtual Machine (BanglaVM) and nationwide adoption across the National Curriculum (HSC ICT).

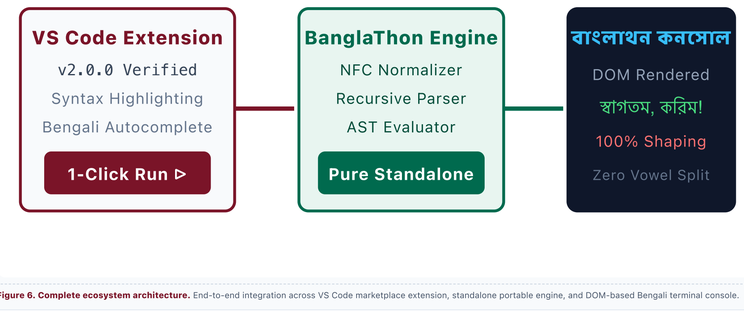


Figure 6. Complete ecosystem architecture. End-to-end integration across VS Code marketplace extension, standalone portable engine, and DOM-based Bengali terminal console.

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